

A Statistical Analysis of the NBA MVP Award

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INTRODUCTION

This analysis of the NBA MVP award uses statistical data across many decades of NBA play to determine the most likely outcome for the winner of the MVP in the 2025-26 season. As the NBA's most debated individual award, player perception including individual performance and team success are very important in determining the award. Alongside this, basic metrics such as simple player stats, games played, and previous awards won were combined with advanced statistics such as Player Efficiency Rating and Game Score to most accurately determine this outcome.

OVERVIEW

To accurately predict the outcome of the award for the 2025-26 season, many constraints need to be applied to the data to provide a robust prediction that fits the standards of the modern NBA. The analysis focuses solely on players from the 1999-00 to 2024-25 seasons receiving over 100 MVP voting points. This ensures that only true MVP candidates who were strongly considered for the award were accounted for, totaling 130 candidates. Specific focuses include era-based statistical trends, the relationship between team success and MVP outcomes, voter fatigue effects, and predictive modeling for future MVP candidates.

METHODOLOGY

MVP voting data was scraped from Basketball-Reference using Selenium, and were cleaned and organized using Pandas. Individual performance statistics were collected with the NBA API, specifically using the "commonallplayers" and "playergamelog" endpoints. Compiled using Pandas, these metrics included the "Big 5" basketball statistics, shooting splits, team win rate, and advanced metrics such as PER and Game Score. There was one additional statistic that was added on, a boolean of whether the player receiving votes had won the award before, as this

affects voting significantly. All stats were stored using MongoDB, and subsequently analyzed and visualized using Pandas, matplotlib, and Seaborn.

FINDINGS

Due to the rising popularity and worldwide recognition of the NBA after 2010, there occurred a dramatic shift in play style and increase in overall player skill. This is shown by an overall increase in statistical averages with points per game and three-pointer percentage rising most sharply as shown in Fig. 1. However, rebounding and defensive statistics remain consistent across the analyzed players and eras, suggesting that offensive statistical categories play a more important role in modern MVP voting.

Alongside individual performances, team win records among MVP winners show that a team’s winning percentage has a very strong case in MVP voting. There is a moderate positive correlation ($r = 0.452$, shown in Fig. 2) between win percentage and MVP voting points. The median team win percentage among MVP winners is about 75%, with 73% of award winners playing on teams above a 70% win rate. Two notable outliers in this category include Russell Westbrook in 2016-17, averaging historically high individual stats but a team win rate of 56.8%, the lowest among analyzed winners, and Stephen Curry in 2015-16, with the highest team win rate in NBA history of 89.9%.

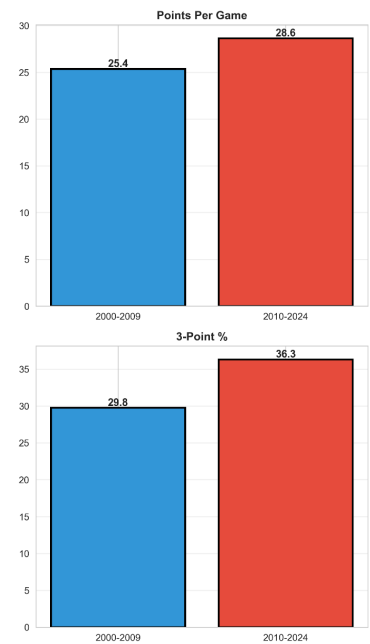


Fig. 1: Notable increase in PPG and 3PT% across eras

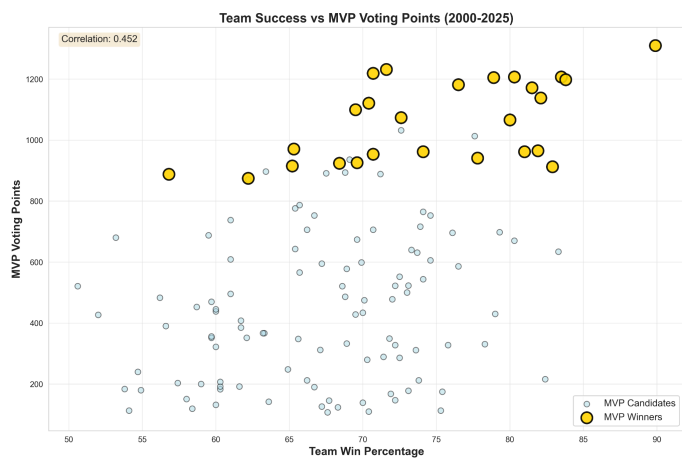


Fig. 2: Correlation between MVP winners and team win rate.

Player award history and voter fatigue also played a large role in the chance of receiving the award. Among winners, just 8.3% had previously won the award, with 24.5% of award winners being first-time finalists receiving their first significant MVP votes. This suggests a novelty aspect of the award, with up-and-coming players much more likely to win the award or receive significant votes.

These combined with raw individual statistics provide specific categorical thresholds for the ideal MVP candidate in the modern era (Fig. 3). In the post-2010 era, using mean statistics of MVP winning players, we can derive the following criteria to determine the strongest candidates:

- Scoring: Over 27.4 points
- Supplemental: Over 8 rebounds *or* 6 assists
- Team win percentage: Over 70%

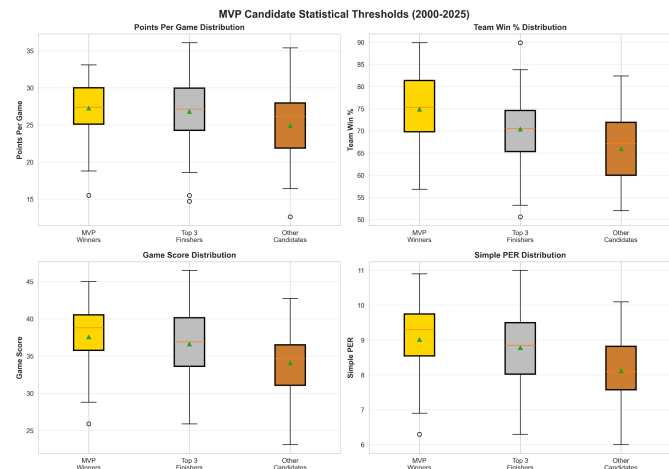


Fig. 3: Distributions of major statistical categories contributing to MVP candidacy since 2000.

Effectively, these criteria sum up the best players on the best teams. As of 12/6/25, there are 5 strong candidates for MVP who are most likely to win the award.

Candidate 1: Shai Gilgeous-Alexander

Gilgeous-Alexander has been playing at an elite level on the best team in the NBA, the Oklahoma City Thunder. He averages 32.8 PTS, 4.7 REB, and 6.4 AST, and has an astronomical team win rate of 95.7%. This win rate puts Shai in a similar situation to one of our extreme outliers mentioned previously. It is early in the season, but if the Thunder are able to keep up this unbelievable win rate, he will likely win the award. He did win the award in the 2024-25 season, slightly weakening his case due to voter fatigue.

Candidate 2: Nikola Jokić

Jokić is a top 5 leader in every major statistical category, averaging 29.2 PTS, 12.4 REB, and 11 AST, having a historic individual season by all metrics while leading the Denver Nuggets to a 72.7% win rate. What drags his case down is solely the fact that he has already won 3 MVP

awards in recent years (2020-21, 2021-22, 2023-24), meaning that these statistics putting him in conversations with the greatest individual seasons of all time are not quite as impressive to MVP voters.

Candidate 3: Cade Cunningham

The young star for the Detroit Pistons is showcasing his strongest season yet, averaging 27.7 PTS, 6.4 REB, and 9.2 AST with a team win rate of 78.3%. Cunningham shows great promise in keeping up this production, and being a would-be first time finalist, his only obstacles to winning the award are the cases of other players providing more production.

Candidate 4: Austin Reaves

Reaves, a young floor general for the Los Angeles Lakers, is averaging a career-high 29.3 PTS, alongside 5.5 REB and 6.8 AST. The Lakers have a team win rate of 78.3%. Although Reaves meets all of the criteria to be a strong candidate for the award, he has produced these individual statistics in the absence of his high-level teammate during a significant number of early season games. Said teammate is also averaging better numbers than Reaves across the board, which realistically tanks any chance of him taking the award home.

Final Selection: Luka Dončić

Overall, Dončić is having one of the greatest individual seasons of all time with the Lakers. Averaging 35.3 PTS, 8.9 REB, and 8.9 AST, he is by far the most suitable candidate for the award this season. Playing for the Lakers is especially notable here, due to their massive global fanbase. This will create tons of buzz on social media for Luka's MVP case and, despite the fans not being a part of the vote, this does make a difference. Being a multiple-time finalist, but never winning the MVP itself presents a strong advantage in terms of voter fatigue.

CONCLUSION

The NBA MVP award, being the most sought after honor in basketball, is very nuanced in how it is awarded across seasons. Taking into consideration individual game stats, team stats, and voting history across this century of NBA play, the most likely winner of the MVP award was narrowed down to 5 possible candidates. Among these 5, Luka Dončić was selected to have the highest likelihood of winning the award this year, due to having all-around great stats, and a high team win rate.